MAT 181 Chapter 8 Practice Problems-1-Ans
Name_____

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question. Express the null hypothesis and the alternative hypothesis in symbolic form. Use the correct symbol (μ, p, σ) for the indicated parameter.

1) An entomologist writes an article in a scientific journal which claims that fewer than 7 in ten thousand male fireflies are unable to produce light due to a genetic mutation. Use the parameter p, the true proportion of fireflies unable to produce light. 1) _____

A) H_0: $p = 0.0007$ H_1: $p > 0.0007$
B) H_0: $p = 0.0007$ H_1: $p < 0.0007$
C) H_0: $p > 0.0007$ H_1: $p \le 0.0007$
D) H_0: $p < 0.0007$ H_1: $p \ge 0.0007$

2) Carter Motor Company claims that its new sedan, the Libra, will average better than 26 miles per gallon in the city. Use μ, the true average mileage of the Libra. 2) _____

A) H_0: $\mu > 26$ H_1: $\mu \le 26$
B) H_0: $\mu < 26$ H_1: $\mu \ge 26$
C) H_0: $\mu = 26$ H_1: $\mu > 26$
D) H_0: $\mu = 26$ H_1: $\mu < 26$

3) A cereal company claims that the mean weight of the cereal in its packets is at least 14 oz. 3) _____

A) H_0: $\mu > 14$ H_1: $\mu \le 14$
B) H_0: $\mu = 14$ H_1: $\mu > 14$
C) H_0: $\mu = 14$ H_1: $\mu < 14$
D) H_0: $\mu < 14$ H_1: $\mu \ge 14$

4) A psychologist claims that more than 4.1 percent of the population suffers from professional problems due to extreme shyness. Use p, the true percentage of the population that suffers from extreme shyness. 4) _____

A) H_0: $p < 4.1\%$ H_1: $p \ge 4.1\%$
B) H_0: $p = 4.1\%$ H_1: $p > 4.1\%$
C) H_0: $p = 4.1\%$ H_1: $p < 4.1\%$
D) H_0: $p > 4.1\%$ H_1: $p \le 4.1\%$

Assume that the data has a normal distribution and the number of observations is greater than fifty. Find the critical z value used to test a null hypothesis. 5) _____

5) $\alpha = 0.05$ for a two-tailed test.

A) ±2.575 B) ±1.764 C) ±1.96 D) ±1.645

6) $\alpha = 0.09$ for a right-tailed test. 6) _____

A) 1.96 B) 1.34 C) ±1.96 D) ±1.34

7) $\alpha = 0.05$ for a left-tailed test. 7) _____

A) ±1.645 B) -1.96 C) -1.645 D) ±1.96

SHORT ANSWER. Write the word or phrase that best completes each statement or answers the question.

Find the value of the test statistic z using $z = \frac{\hat{p} - p}{\sqrt{\frac{pq}{n}}}$ **.**

8) A claim is made that the proportion of children who play sports is less than 0.5, and the sample statistics include $n = 1671$ subjects with 30% saying that they play sport. 8) _____

9) The claim is that the proportion of accidental deaths of the elderly attributable to residential falls is more than 0.10, and the sample statistics include $n = 800$ deaths of the elderly with 15% of them attributable to residential falls.

9) _____

A) 4.71 B) -4.71 C) 3.96 D) -3.96

Use the given information to find the P-value. Also, use a 0.05 significance level and state the conclusion about the null hypothesis (reject the null hypothesis or fail to reject the null hypothesis).

10) The test statistic in a right-tailed test is z = 0.52. 10) _____

A) 0.0195; reject the null hypothesis
B) 0.3015; reject the null hypothesis
C) 0.6030; fail to reject the null hypothesis
D) 0.3015; fail to reject the null hypothesis

11) The test statistic in a left-tailed test is z = -1.83. 11) _____

A) 0.0672; reject the null hypothesis
B) 0.9664; fail to reject the null hypothesis
C) 0.0336; reject the null hypothesis
D) 0.0672; fail to reject the null hypothesis

Formulate the indicated conclusion in nontechnical terms. Be sure to address the original claim.

12) Carter Motor Company claims that its new sedan, the Libra, will average better than 30 miles per gallon in the city. Assuming that a hypothesis test of the claim has been conducted and that the conclusion is to reject the null hypothesis, state the conclusion in nontechnical terms. 12) _____

A) There is sufficient evidence to support the claim that the mean is less than 30 miles per gallon.
B) There is not sufficient evidence to support the claim that the mean is less than 30 miles per gallon.
C) There is not sufficient evidence to support the claim that the mean is greater than 30 miles per gallon.
D) There is sufficient evidence to support the claim that the mean is greater than 30 miles per gallon.

13) A skeptical paranormal researcher claims that the proportion of Americans that have seen a UFO, p, is less than 2 in every ten thousand. Assuming that a hypothesis test of the claim has been conducted and that the conclusion is failure to reject the null hypothesis, state the conclusion in nontechnical terms. 13) _____

A) There is sufficient evidence to support the claim that the true proportion is less than 2 in ten thousand.
B) There is sufficient evidence to support the claim that the true proportion is greater than 2 in ten thousand.
C) There is not sufficient evidence to support the claim that the true proportion is greater than 2 in ten thousand.
D) There is not sufficient evidence to support the claim that the true proportion is less than 2 in ten thousand.

Assume that a hypothesis test of the given claim will be conducted. Identify the type I or type II error for the test.

14) The principal of a school claims that the percentage of students at his school that come from single-parent homes is 13%. Identify the type II error for the test. 14) _____

A) Fail to reject the claim that the percentage of students that come from single-parent homes is equal to 13% when that percentage is actually different from 13%.
B) Reject the claim that the percentage of students that come from single-parent homes is equal to 13% when that percentage is actually less than 13%.
C) Reject the claim that the percentage of students that come from single-parent homes is equal to 13% when that percentage is actually 13%.
D) Fail to reject the claim that the percentage of students that come from single-parent homes is equal to 13% when that percentage is actually 13%.

15) A medical researcher claims that 10% of children suffer from a certain disorder. Identify the type I error for the test. 15) _____

A) Reject the claim that the percentage of children who suffer from the disorder is equal to 10% when that percentage is actually 10%.
B) Fail to reject the claim that the percentage of children who suffer from the disorder is equal to 10% when that percentage is actually different from 10%.
C) Fail to reject the claim that the percentage of children who suffer from the disorder is equal to 10% when that percentage is actually 10%.
D) Reject the claim that the percentage of children who suffer from the disorder is different from 10% when that percentage really is different from 10%.